

Choosing an ingestion method

The decision sequence, the four criteria the exam guide names, and eight worked choices.

Name the **source type** first. It eliminates most options before you have thought about anything else. The four criteria then break the tie.

The decision sequence

01

Name the source

Files, database, SaaS, message bus, or API.

›

02

Does a managed connector exist?

If yes, and it meets the requirements, stop here.

›

03

Check volume and frequency

These separate COPY INTO from Auto Loader, and batch from streaming.

›

04

Check governance

Unity Catalog, credentials, audit. Often eliminates an otherwise good option.

›

05

Drop a layer only if forced

Custom code is the answer when a requirement demands it, not by default.

WORK IT IN THIS ORDER

Five steps in order. Most exam questions are decided by step one or step two.

The four criteria, named in the exam guide

CRITERION	WHAT IT DECIDES
Data volume	Thousands of files means COPY INTO. Millions means Auto Loader.
Ingestion frequency	Daily batch, bursts through the day, or continuous seconds-level.
Data types	Files, tables, semi-structured payloads, or unstructured documents.
Governance needs	Unity Catalog grants, credential handling, audit and lineage.

ON THE EXAM

These are the guide's own words. When a scenario states a number or a frequency, it is handing you one of these criteria.

Criterion one: data volume

- ◆ **Thousands of files over time:** COPY INTO is appropriate and simpler.
- ◆ **Millions or more:** Auto Loader, because discovery cost is what breaks first.
- ◆ **Row count is not the criterion. File count is.** A billion rows in twelve Parquet files is not a volume problem.
- ◆ **Above millions per hour:** Auto Loader with file notification rather than directory listing.

ON THE EXAM

The distinction that catches people: many small files is a discovery problem, while a lot of data is a processing problem. Different fixes.

Criterion two: ingestion frequency

- ◆ **Once or a few times a day:** a scheduled job, with COPY INTO or Auto Loader using AvailableNow.
- ◆ **Bursts through the day at unpredictable times:** a file arrival trigger, or serverless to avoid idle cost.
- ◆ **Continuous, seconds matter:** a genuine stream, with a cluster that stays up.
- ◆ **Continuous costs continuously.** Only choose it when latency is a stated requirement, not an aspiration.

ON THE EXAM

'Near real time' in a scenario usually means minutes. Read the stated latency before assuming a continuous stream is required.

Criterion three: data types

- ◆ **Structured files** in object storage: COPY INTO or Auto Loader.
- ◆ **Semi-structured JSON with drift**: Auto Loader, for schema evolution and the rescued data column.
- ◆ **Unstructured documents** such as PDFs or images: a volume, and possibly a file source connector.
- ◆ **Message bus payloads**: Structured Streaming or a streaming connector. Never Auto Loader.

ON THE EXAM

Auto Loader reads files from object storage. It does not read Kafka. That single fact settles a whole family of questions.

Criterion four: governance

- ◆ **A hardcoded credential is disqualifying**, whatever else the option does well.
- ◆ **A DBFS mount bypasses Unity Catalog entirely**. Deprecated, and wrong wherever governance is mentioned.
- ◆ **Destination tables must be Unity Catalog governed** for lineage and audit to work.
- ◆ **Managed connectors store credentials as a UC connection**, which is a governed securable rather than a string.

ON THE EXAM

Governance is the criterion most likely to eliminate a technically correct answer. Check it last, but do check it.

The whole of domain 2 on one slide

METHOD	SOURCE	BEST WHEN
COPY INTO	Files in object storage	Thousands of files, SQL-first, stable schema
Auto Loader	Files in object storage	Millions of files, schema drift, continuous arrival
Streaming tables	Files, in SQL	The recommended SQL route; wraps Auto Loader
Managed connector	SaaS, databases, enterprise files	A connector exists for your source
Streaming connector	Message buses	Kafka, Pub/Sub, Kinesis and similar
JDBC or REST in a notebook	Databases and APIs	No connector fits, or a specific query is needed
Partner connector	Long tail of SaaS sources	Databricks has no connector but a partner does
Lakehouse Federation	Databases	No copy should exist at all

ON THE EXAM

Eight methods. Read the middle column first — the source type alone eliminates most rows.

Partner connectors

- ◆ **Partner Connect is a portal** of validated data, analytics and AI tools that integrate with Databricks.
- ◆ **Fivetran is the canonical ingestion partner**, offering connectors to a very long list of SaaS sources.
- ◆ **They connect through a SQL warehouse or cluster**, and write into Delta tables in your workspace.
- ◆ **The reason to use one is coverage.** A partner may support a source Databricks does not.

ON THE EXAM

The exam guide names partner connectors as one of the options to prioritise between. Coverage is what they add.

Instant eliminations

Patterns that make an option wrong before you evaluate it properly

- ◆ **Auto Loader or COPY INTO against a message bus.** Wrong source type.
- ◆ **A password written into a notebook.** Governance failure.
- ◆ **A DBFS mount or the DBFS root.** Deprecated and ungoverned.
- ◆ **Deleting a checkpoint to reprocess.** Causes mass duplication.
- ◆ **An invented limit**, such as a maximum table size or a fixed retention period.

ON THE EXAM

Learning to eliminate is faster than learning to select. Two eliminations often leave one plausible answer and one obvious one.

Two ways to get it wrong

OVER-ENGINEERING

Machinery nobody needed

- A streaming pipeline for a quarterly reference file
- File notification for a directory of forty files
- A custom connector where a managed one exists
- Costs money and adds things that can break

UNDER-ENGINEERING

Something that will not hold

- A plain directory read for a growing feed
- No incremental state, so duplicates on re-run
- No schema handling for a source that drifts
- Works today, fails as volume or time increases

BOTH ARE TESTED

The exam tests both. An option can be wrong for doing too much just as easily as for doing too little.

Tiebreakers for the whole domain

WHEN A QUESTION SAYS	REACH FOR
Thousands of files, SQL team	COPY INTO
Millions of files, or schema drift	Auto Loader
Salesforce, Workday, SQL Server	A managed connector
Kafka, Kinesis, Pub/Sub	Structured Streaming or a streaming connector
A niche SaaS app Databricks does not cover	A partner connector
A query no connector can express	JDBC in a notebook
The data must not be copied	Lakehouse Federation
A 40-row file maintained by hand	A volume. No machinery at all.

ON THE EXAM

Eight lines for eight methods. Now eight scenarios to practise applying them under pressure.

A finance team receives about 30 CSV files each night from three banks, into a Unity Catalog volume. The schema has not changed in four years and is contractually fixed. The team writes SQL and has no Python experience.

They need the files loaded into a bronze table each morning, without reloading what was already loaded.

Which method fits best?

- ☐ A Auto Loader with file notification mode and schema evolution
- ☐ B COPY INTO on a scheduled job
- ☐ C A nightly spark.read over the directory, overwriting the table
- ☐ D A managed connector for each of the three banks

◆ PAUSE AND DECIDE

Which method fits best?

- A

 Auto Loader with file notification
Every part of it is aimed at a problem this team does not have. Thirty files a night is not a discovery problem, and the schema is contractually fixed.
- B

 COPY INTO on a schedule
Correct. Thousands of files is COPY INTO's range and this is thirty a night. It is one SQL statement, idempotent by default, and the team writes SQL.
- C

 spark.read and overwrite nightly
No incremental state, so runtime grows with history forever. This is the module 3 anti-pattern.
- D

 A managed connector per bank
The banks deliver files. Managed connectors read SaaS applications and databases, not a directory someone drops CSVs into.

THE SIGNAL

Three details point the same way: 30 files, a fixed schema, and a SQL-only team. Every one of them favours the simpler tool.

A logistics company needs data from a niche freight-tracking SaaS platform. Databricks has no managed connector for it. The platform has a documented REST API with OAuth, pagination and rate limits, and changes its API roughly twice a year.

The data team is two people. Their manager asks for the lowest-maintenance option that still lands governed tables.

What should they do?

- ☐ A Write a Python notebook that calls the REST API and handles OAuth, pagination and rate limits
- ☐ B Check Partner Connect for a partner connector that supports the platform
- ☐ C Use Auto Loader against the platform's API endpoint
- ☐ D Use Lakehouse Federation to query the platform in place

◆ PAUSE AND DECIDE

What should they do?

- A

 Write the REST notebook
Viable, and the fallback if B fails. But an API that changes twice a year means recurring breakage for a two-person team. Highest maintenance, not lowest.
- B

 Check Partner Connect first
Correct. Partner connectors exist precisely for sources Databricks has not covered, and a partner absorbs the API changes rather than your two engineers.
- C

 Auto Loader against the API
Wrong source type. Auto Loader reads files in object storage; it cannot call a REST endpoint.
- D

 Lakehouse Federation
Federation connects to databases as foreign catalogs, not to arbitrary SaaS REST APIs.

THE SIGNAL

'No managed connector exists' does not mean 'write it yourself'. Partner connectors sit between those two, and the guide names them.

A retailer publishes clickstream events to an Apache Kafka topic at around 40,000 messages per second. They need those events in a bronze Delta table with a latency of under a minute. An engineer suggests configuring Auto Loader with file notification mode, since it is designed for high-volume ingestion.

How should the team respond?

- ☐ A Agree; Auto Loader with file notification handles this volume comfortably
- ☐ B Auto Loader reads files from object storage and cannot read Kafka; use Structured Streaming or a streaming connector
- ☐ C Agree, but use directory listing mode instead for lower latency
- ☐ D Use COPY INTO with a short schedule to poll the topic

◆ PAUSE AND DECIDE

How should the team respond?

- A

 Auto Loader handles this volume
Volume is not the objection. Auto Loader's source is a directory in object storage, and Kafka is not one.
- B

 Wrong source type; use Structured Streaming or a streaming connector
Correct. Message buses need a streaming source. Auto Loader's cloudFiles source reads files, not topics.
- C

 Directory listing for lower latency
Accepts the same false premise and then argues about a setting. There is no directory to list.
- D

 COPY INTO polling the topic
COPY INTO reads files from object storage. It has no mechanism for subscribing to a topic.

THE SIGNAL

The engineer's reasoning is about volume, which is a real strength of Auto Loader. It is irrelevant here, because the source type rules it out first.

A media company lands roughly 900,000 JSON files a day into cloud storage from twelve partner feeds. The partners add fields without notice, several times a year. The pipeline supports a live dashboard and must not stop.

The current process is a COPY INTO statement scheduled hourly. It has begun failing whenever a partner adds a field.

What should replace it?

- ☐ A COPY INTO with mergeSchema set to true in COPY_OPTIONS
- ☐ B Auto Loader with schemaEvolutionMode set to rescue
- ☐ C Auto Loader with schemaEvolutionMode set to failOnNewColumns
- ☐ D A managed connector for each of the twelve partners

◆ PAUSE AND DECIDE

What should replace it?

- A

COPY INTO with mergeSchema

Fixes the schema failure but leaves 900,000 files a day being discovered by directory listing. Solves one problem, keeps the bigger one.
- C

Auto Loader with failOnNewColumns

Right tool, wrong mode. It stops the pipeline on every new field, which is exactly what the must-not-stop requirement forbids.
- B

Auto Loader with rescue mode

Correct. Auto Loader handles the file volume, and rescue keeps the stream running while capturing new fields for later review.
- D

Twelve managed connectors

The partners deliver files to object storage. Managed connectors read SaaS applications and databases, not partner file drops.

THE SIGNAL

Two requirements, and A satisfies only one. Volume points to Auto Loader; 'must not stop' then picks the mode.

An insurer needs policy changes from an on-premises SQL Server, including deletes, so that a fraud model sees the full change history. The database sits behind a corporate firewall with no public route.

The platform team has an ExpressRoute link already in place and wants credentials governed rather than stored in code.

What is the right approach?

- ☐ A A JDBC notebook reading the table nightly, with credentials in a secret scope
- ☐ B A managed database connector using CDC, connecting over the existing ExpressRoute link
- ☐ C A query-based connector querying the table on a schedule
- ☐ D Lakehouse Federation, so no copy is made

◆ PAUSE AND DECIDE

What is the right approach?

- A

Nightly JDBC with secret scope
Governance is handled, but a nightly snapshot cannot see deletes or intermediate changes. It shows the state at query time only.
- C

Query-based connector on a schedule
Governed and simpler, but it has the same blind spot as A. Querying sees current state, not the changes between runs.
- B

Managed CDC connector over ExpressRoute
Correct. CDC captures every change including deletes, the connection is a governed UC securable, and ExpressRoute is a documented path for on-premises sources.
- D

Lakehouse Federation
Makes no copy, so there is no history to give the fraud model, and every query would traverse the firewall to a production database.

THE SIGNAL

'Including deletes' is the deciding phrase. Only change data capture sees a row that no longer exists.

A team is decommissioning an old analytics platform. They must load 4 TB of historical Parquet, sitting in about 1,100 files in cloud storage, into a Delta table. This is a one-off migration. Nothing further will ever be written to that location.

An architect proposes building an Auto Loader stream with a checkpoint and file notification mode.

What is the best response?

- ☐ A Agree; Auto Loader is the standard tool for ingesting files from cloud storage
- ☐ B A one-off load of 1,100 files needs neither a checkpoint nor notification; COPY INTO or a batch read is sufficient
- ☐ C Use Auto Loader but with directory listing rather than file notification
- ☐ D Split the files across twelve separate Auto Loader streams to parallelise

◆ PAUSE AND DECIDE

What is the best response?

- A

 Agree, Auto Loader is standard
Standard for ongoing ingestion. A checkpoint tracks progress across runs, and there will never be a second run.
- C

 Auto Loader with directory listing
Better than file notification here, but still builds a checkpoint and a stream for something that happens exactly once.
- B

 COPY INTO or a batch read is sufficient
Correct. 1,100 files is well inside COPY INTO's range, and the incremental machinery has no future to serve.
- D

 Twelve parallel streams
Twelve checkpoints instead of one, for a load that is not discovery-bound. It multiplies the over-engineering.

THE SIGNAL

4 TB sounds large, and it is not the criterion. 1,100 files is the number that matters, and the load happens once.

A hospital's analysts need to join patient demographics, held in a clinical PostgreSQL system, against treatment outcomes already in the lakehouse. The clinical system is subject to a data protection agreement that prohibits copying patient records outside it.

Query volumes would be low, a few dozen analyst queries a day.

What should the team use?

- ☐ A A managed PostgreSQL CDC connector into a governed bronze table
- ☐ B A JDBC notebook writing demographics into a restricted catalog
- ☐ C Lakehouse Federation, registering PostgreSQL as a foreign catalog
- ☐ D A query-based connector on a nightly schedule

◆ PAUSE AND DECIDE

What should the team use?

- A

 Managed CDC connector into bronze
Copies the records. Governed and well built, and still a copy, which the agreement prohibits.
- B

 JDBC into a restricted catalog
Restricting access to a copy does not stop it being a copy. Same violation, less automation.
- C

 Lakehouse Federation as a foreign catalog
Correct. Federation queries in place and makes no copy, and low query volume means the load on the clinical system stays acceptable.
- D

 Query-based connector nightly
Query-based describes how the source is read, not whether a copy results. It still writes records into destination tables.

THE SIGNAL

Three of the four options ingest. The constraint is not about governance quality; it is that a copy must not exist at all.

A team has an ingestion pipeline pulling from Salesforce, built and working in their development workspace using the managed connector. It was configured entirely through the UI.

They now need the same pipeline in test and production workspaces, with different destination catalogs per environment. It must be promoted through code review rather than rebuilt by hand.

How should they proceed?

- A Repeat the UI configuration in each workspace, carefully matching the settings
- B Define the pipeline in a Declarative Automation Bundle with per-target variables, and deploy with the CLI
- C Export the pipeline as JSON and import it into each workspace manually
- D Use one shared pipeline in development and grant the other workspaces access to its output

◆ PAUSE AND DECIDE

How should they proceed?

- A

 Repeat the UI configuration
Works once, drifts immediately, and satisfies neither the code review requirement nor reproducibility.
- B

 A bundle with per-target variables, deployed by CLI
Correct. Bundles are the documented way to deploy ingestion pipelines across dev, test and prod with source control and CI/CD.
- C

 Export and import JSON manually
Closer, but the manual step is the problem. Nothing is under review, and the per-environment differences are still hand-edited.
- D

 One shared pipeline in development
Production would depend on a development workspace. It also does not give test and prod their own destination catalogs.

THE SIGNAL

Ingestion pipelines are deployable assets. The Lakeflow Connect docs name Declarative Automation Bundles for exactly this.

Name the source. Take the most managed option that meets every stated requirement. **Let the stated requirements, not your instincts, force you down a layer.**